

Eight edge cases on the A10-deterministic board

This is the A10 assignment's own soccer Markov game: a 7×5 field, goal rows {1, 2, 3}, four actions per player (U/D/L/R), and fully deterministic move resolution — every joint action maps to exactly one outcome, never a coin flip. This project's headline result on that board is that every one of its 2,380 legal states resolves to a *pure* equilibrium. But "pure" is not the same as "obvious": a collision rule where winning a contested square still hands the ball to whoever lost it, a boundary rule where a move past the edge holds in place everywhere *except* a goal row (where it scores instead), and outright ties between several equally-good actions all hide real structure that a single printed action can't show. These eight cases, found by directly searching the exact solve for the interesting shapes, put that structure on a board — the same ones live in the [board explorer](#).

Ties, not tie-breaking. A pure saddle's (i^*, j^*) is just the *first* row/column achieving the guaranteed value — when more than one does, every one of them is equally optimal. The board explorer used to print only the first and imply it was the answer; it no longer does. Every case below prints the **full tied set**, the solver's own one-hot (tie-broken) policy, and the **uniform-over-ties display policy** the site now actually shows and actually plays in Step/Play/ Simulate — a uniform mix over a saddle's whole tied set is itself still an optimal strategy, not a display approximation.

Hold vs. score. A move that runs into the board edge is a genuine no-op everywhere *except* one place: the ball carrier, in a goal row, pushing past their own attacking edge, which ends the game instead of holding in place. Two cases below (4 and 5) exist specifically to put that on a diagram.

The collision rule itself. Winning the race to a contested cell does not mean keeping the ball — the loser of a contest or a swap gets it regardless of who reached the square first. Every transition on this page is read straight from `game.transitions()`, not asserted.

Board	7×5, goal rows {1, 2, 3}, move_order="deterministic" – the explorer's "canonical_det"
Rows / Columns	Player 0 / player 1, fixed for every state, never reoriented by who has the ball
Payoff	Player 0's Q-value
 mark	wall-clamped hold: legal to pick, but there's no cell there, so it does nothing
 mark	the carrier, in a goal row, pushing past their own edge: this scores, it is not a hold

Case 1 (4, 3, 5, 3, 0) – The swap trap

Player 0's U/D/L never touch player 1 at all this turn, so all three guarantee 0 regardless of player 1's reply — only R reaches player 1's square, and against L it triggers a genuine swap $((R, L) \rightarrow (5, 3, 4, 3, 1))$, verified below) that hands player 1 the ball, dragging R's own worst case to -0.59 . Player 1's unique safe column is R for an unrelated reason: U/D/L each let a smart player

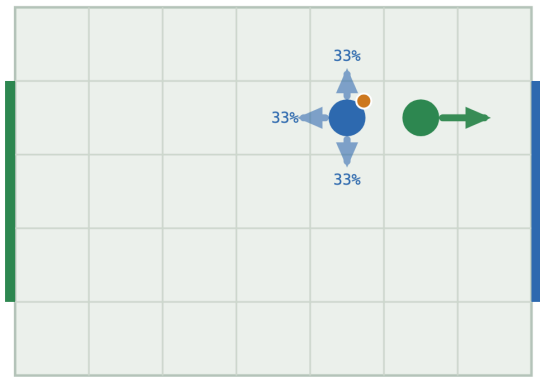
0 reach a good, independent square elsewhere in the matrix (0.81 / 0.81 / 0.729) that has nothing to do with the swap cell.

Type: pure saddle, $V = +0.000000$ — player 0's maximin ties at U/D/L (not unique); player 1's minimax is uniquely R.

Solver's own policy (tie-broken): $p_0 = U\ 100\% \cdot p_1 = R\ 100\%$

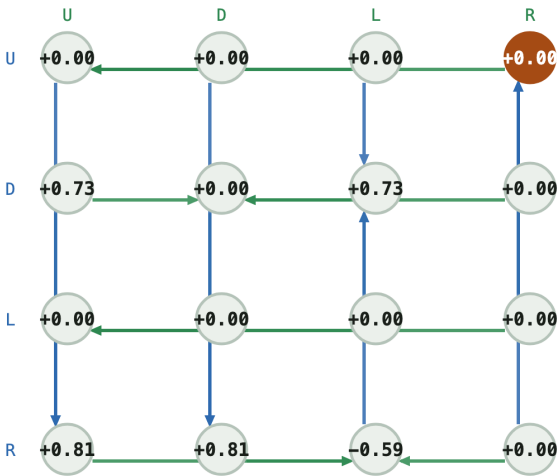
Display policy (uniform over ties, what the site shows and plays): $p_0 = U\ 33.3\% / D\ 33.3\% / L\ 33.3\% \cdot p_1 = R\ 100\%$

The swap trap



pure equilibrium · $V = +0.000$

Q matrix -- (4, 3, 5, 3, 0)



pure saddle at U/R

Left: where the players are, every tied action drawn as an equal-weight arrow. Right: the full 4×4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.729000	0.000000	0.729000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.810000	0.810000	-0.590490	0.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x0, y0, x1, y1, b).

P0 \ P1	U	D	L	R
U	(4,4,5,4,0)	(4,4,5,2,0)	(4,4,4,3,0)	(4,4,6,3,0)
D	(4,2,5,4,0)	(4,2,5,2,0)	(4,2,4,3,0)	(4,2,6,3,0)
L	(3,3,5,4,0)	(3,3,5,2,0)	(3,3,4,3,0)	(3,3,6,3,0)
R	(5,3,5,4,0)	(5,3,5,2,0)	(5,3,4,3,1) – swap, ball to loser	(5,3,6,3,0)

Case 2 (0, 0, 0, 2, 0) – Pinned in the corner

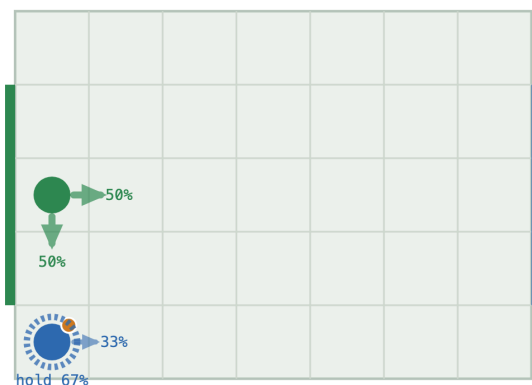
Player 0's U is the one risky action: it targets the empty cell (0,1), which player 1's D also targets. Neither player is standing there, so it's an *open* (non-blocked) contest — player 0 physically wins the race ((U, D) → (0,1,0,2,1), verified below) but the A10 rule still gives the ball to the loser regardless of who reached the cell first, crashing the value to -0.9. D, L, and R never reach that cell, so they tie safely at 0.

Type: pure saddle, $V = +0.000000$ — player 0's maximin ties at D/L/R (not unique); player 1's minimax ties at D/R (not unique).

Solver's own policy (tie-broken): $p_0 = D \ 100\% \cdot p_1 = D \ 100\%$

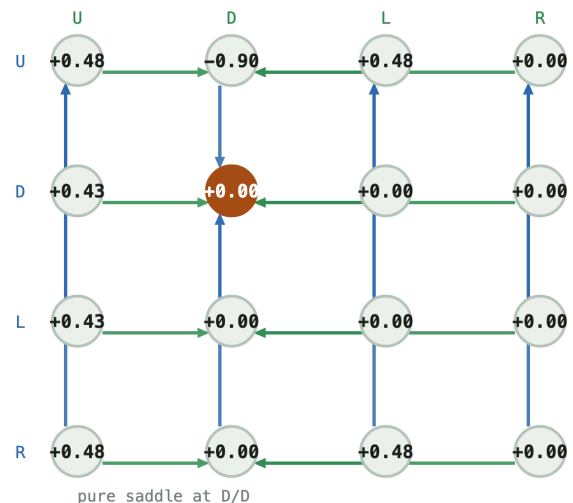
Display policy (uniform over ties, what the site shows and plays): $p_0 = D \ 33.3\% / L \ 33.3\% / R \ 33.3\% \cdot p_1 = D \ 50\% / R \ 50\%$

Pinned in the corner



pure equilibrium · $V = +0.000$

Q matrix -- (0, 0, 0, 2, 0)



Left: where the players are, every tied action drawn as an equal-weight arrow. Right: the full 4×4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.000000	0.000000
L	0.430467	0.000000	0.000000	0.000000
R	0.478297	0.000000	0.478297	0.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x0, y0, x1, y1, b).

P0 \ P1	U	D	L	R
U	(0,1,0,3,0)	(0,1,0,2,1) — open contest, ball to loser	(0,1,0,2,0)	(0,1,1,2,0)
D	(0,0,0,3,0)	(0,0,0,1,0)	(0,0,0,2,0)	(0,0,1,2,0)
L	(0,0,0,3,0)	(0,0,0,1,0)	(0,0,0,2,0)	(0,0,1,2,0)
R	(1,0,0,3,0)	(1,0,0,1,0)	(1,0,0,2,0)	(1,0,1,2,0)

Case 3 (3, 4, 4, 0) — The standoff

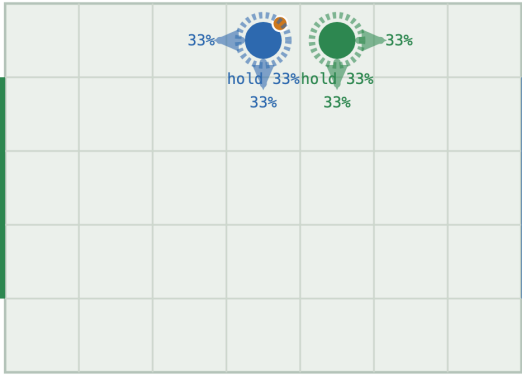
Both players are already on the top row, one cell apart. U is wall-clamped (a genuine no-op) for both — not a real move upward, despite being what the solver happens to print first. R is player 0's one risky action (it reaches player 1's square: blocked against U, a real swap against L); L is player 1's mirror risk. D never touches the opponent for either player, so U/D/L (player 0) and U/D/R (player 1) all tie at the exact same safe value.

Type: pure saddle, $V = +0.000000$ — player 0's maximin ties at U/D/L (not unique); player 1's minimax ties at U/D/R (not unique).

Solver's own policy (tie-broken): $p_0 = U \ 100\% \cdot p_1 = U \ 100\%$

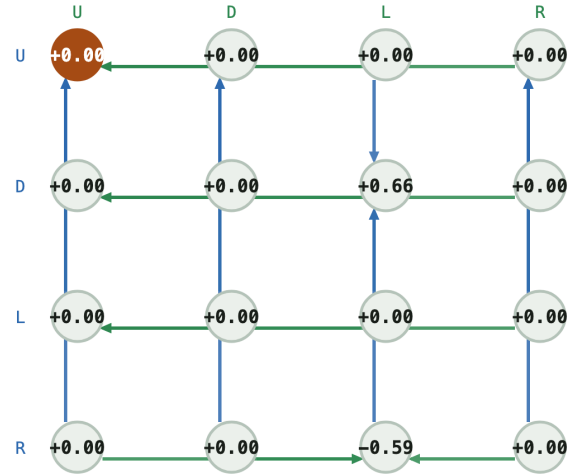
Display policy (uniform over ties, what the site shows and plays): $p_0 = U \ 33.3\% / D \ 33.3\% / L \ 33.3\% \cdot p_1 = U \ 33.3\% / D \ 33.3\% / R \ 33.3\%$

The standoff



pure equilibrium · $V = +0.000$

Q matrix -- (3, 4, 4, 4, 0)



pure saddle at U/U

Left: both players' wall-clamped U shares a dashed hold ring with two real, equal-weight arrows. Right: the full 4x4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.656100	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	-0.590490	0.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x0, y0, x1, y1, b).

P0 \ P1	U	D	L	R
U	(3,4,4,4,0) — both hold	(3,4,4,3,0)	(3,4,4,4,1) — blocked, ball to loser	(3,4,5,4,0)
D	(3,3,4,4,0)	(3,3,4,3,0)	(3,3,3,4,0)	(3,3,5,4,0)
L	(2,4,4,4,0)	(2,4,4,3,0)	(2,4,3,4,0)	(2,4,5,4,0)
R	(3,4,4,4,1) — blocked, ball to loser	(4,4,4,3,0)	(4,4,3,4,1) — swap, ball to loser	(4,4,5,4,0)

Case 4 (6, 1, 4, 1, 0) — The open goal

Player 0 already sits at the right wall in a goal row, with the ball. Off a goal row this exact same boundary push would just hold in place — here it scores instead: `verified, game.transitions()` returns a terminal player-0 win for

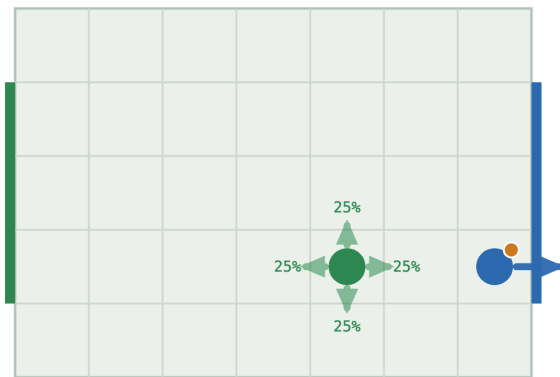
R against every one of player 1's replies. R is the unique optimum (maximin 1.0); player 1's columns all tie at minimax 1.0 too, because there is nothing any of them can do about it.

Type: pure saddle, $V = +1.000000$ — player 0's maximin is uniquely R; player 1's minimax ties at all four (not unique, and moot).

Solver's own policy (tie-broken): $p_0 = R \ 100\% \cdot p_1 = U \ 100\%$

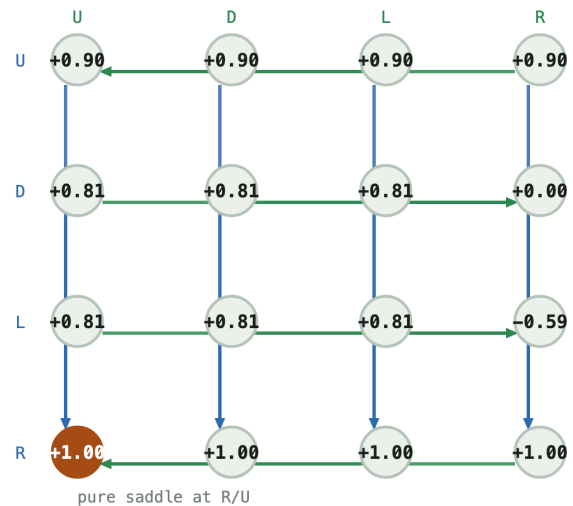
Display policy (uniform over ties, what the site shows and plays): $p_0 = R \ 100\% \cdot p_1 = U \ 25\% / D \ 25\% / L \ 25\% / R \ 25\%$

The open goal



pure equilibrium · $V = +1.000$

Q matrix -- (6, 1, 4, 1, 0)



Left: player 0's R draws as a real solid arrow into the goal strip, not a hold ring — this is the mark, not the mark. Right: the full 4x4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	0.900000	0.900000	0.900000	0.900000
D	0.810000	0.810000	0.810000	0.000000
L	0.810000	0.810000	0.810000	-0.590490
R	1.000000	1.000000	1.000000	1.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x_0, y_0, x_1, y_1, b) or a goal.

P0 \ P1	U	D	L	R
U	(6,2,4,2,0)	(6,2,4,0,0)	(6,2,3,1,0)	(6,2,5,1,0)
D	(6,0,4,2,0)	(6,0,4,0,0)	(6,0,3,1,0)	(6,0,5,1,0)
L	(5,1,4,2,0)	(5,1,4,0,0)	(5,1,3,1,0)	(5,1,4,1,1)
R	player 0 scores	player 0 scores	player 0 scores	player 0 scores

Case 5 (0, 0, 0, 1, 1) – The open net

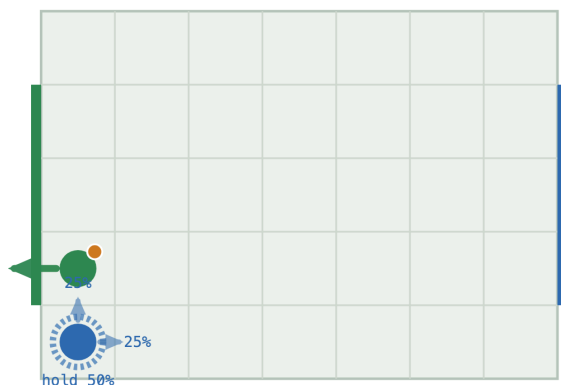
The mirror of the open goal from player 1's side: player 1 is at the left wall in a goal row, ball in hand, and L scores unconditionally (verified against every reply). The bleaker half of this case is player 0's row: every one of U/D/L/R ties at the exact same -1.0 — there is nothing player 0 can do differently that changes the outcome, which is itself worth seeing on a board rather than only in a number.

Type: pure saddle, $V = -1.000000$ — player 0's maximin ties at all four (not unique, and moot); player 1's minimax is uniquely L.

Solver's own policy (tie-broken): $p_0 = U \ 100\% \cdot p_1 = L \ 100\%$

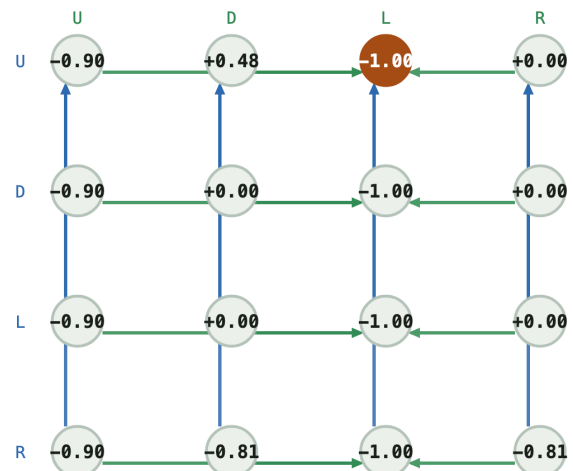
Display policy (uniform over ties, what the site shows and plays): $p_0 = U \ 25\% / D \ 25\% / L \ 25\% / R \ 25\% \cdot p_1 = L \ 100\%$

The open net



pure equilibrium · $V = -1.000$

Q matrix -- (0, 0, 0, 1, 1)



pure saddle at U/L

Left: player 1's L draws as a real solid arrow into the goal strip, not a hold ring; player 0's four equal arrows show a certain loss no matter what. Right: the full 4×4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	-0.900000	0.478297	-1.000000	0.000000
D	-0.900000	0.000000	-1.000000	0.000000
L	-0.900000	0.000000	-1.000000	0.000000
R	-0.900000	-0.810000	-1.000000	-0.810000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x0, y0, x1, y1, b) or a goal.

P0 \ P1	U	D	L	R
U	(0,1,0,2,1)	(0,1,0,0,0)	player 1 scores	(0,1,1,1,1)
D	(0,0,0,2,1)	(0,0,0,1,0)	player 1 scores	(0,0,1,1,1)
L	(0,0,0,2,1)	(0,0,0,1,0)	player 1 scores	(0,0,1,1,1)
R	(1,0,0,2,1)	(1,0,0,0,1)	player 1 scores	(1,0,1,1,1)

Case 6 (0, 1, 0, 3, 0) — The getaway

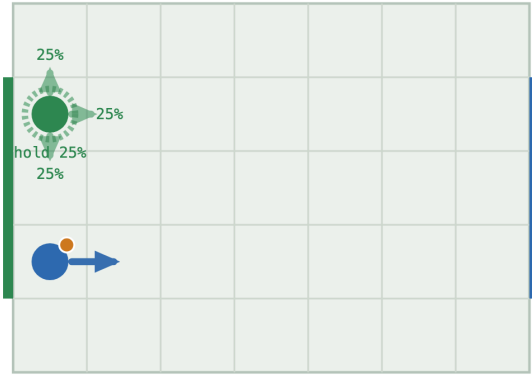
R is player 0's *unique* safe action (0.531441, no tie) — an entirely independent move to (1, 1) that never depends on player 1's reply. U looks just as reasonable but is a trap: against D, both target the same empty cell (0, 2); player 0 wins the race there ((U, D) → (0, 2, 0, 3, 1), verified below) and immediately loses the ball anyway, crashing to -0.9. Player 1's columns all tie at 0.531441 — there is no reply that slows the escape down.

Type: pure saddle, $V = +0.531441$ — player 0's maximin is uniquely R; player 1's minimax ties at all four (not unique, and moot).

Solver's own policy (tie-broken): p0 = R 100% · p1 = U 100%

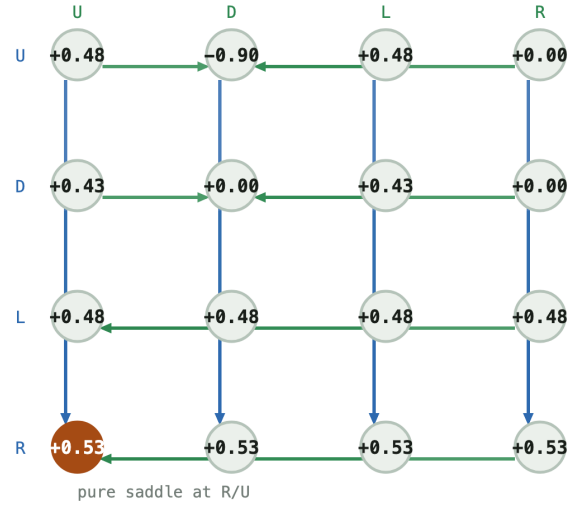
Display policy (uniform over ties, what the site shows and plays): p0 = R 100% · p1 = U 25% / D 25% / L 25% / R 25%

The getaway



pure equilibrium · $V = +0.531$

Q matrix -- (0, 1, 0, 3, 0)



pure saddle at R/U

Left: player 0's single, unique escape arrow against player 1's four indifferent replies.
Right: the full 4x4 Q matrix as a node-and-arrow graph.

	U	D	L	R
U	0.478297	-0.900000	0.478297	0.000000
D	0.430467	0.000000	0.430467	0.000000
L	0.478297	0.478297	0.478297	0.478297
R	0.531441	0.531441	0.531441	0.531441

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x_0, y_0, x_1, y_1, b) .

P0 \ P1	U	D	L	R
U	$(0, 2, 0, 4, 0)$	$(0, 2, 0, 3, 1)$ — open contest, ball to loser	$(0, 2, 0, 3, 0)$	$(0, 2, 1, 3, 0)$
D	$(0, 0, 0, 4, 0)$	$(0, 0, 0, 2, 0)$	$(0, 0, 0, 3, 0)$	$(0, 0, 1, 3, 0)$
L	$(0, 1, 0, 4, 0)$	$(0, 1, 0, 2, 0)$	$(0, 1, 0, 3, 0)$ — both hold	$(0, 1, 1, 3, 0)$
R	$(1, 1, 0, 4, 0)$	$(1, 1, 0, 2, 0)$	$(1, 1, 0, 3, 0)$	$(1, 1, 1, 3, 0)$

Case 7 (1, 0, 0, 1, 0) — The tightest tie

No state on this entire board ever has a fully unique best action for *both* players simultaneously — checked exhaustively across all 2,380 states, the shape $(1, 1)$ never occurs. This is the smallest tie shape that does occur

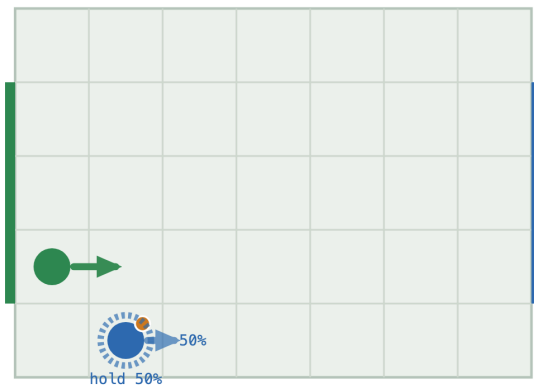
(only 16 of 2,380 states): player 0's D and R tie exactly; player 1's R is uniquely optimal. U and L each look just as safe as D/R, but each hides its own contest: (U, R) both target the same cell (1,1), and (L, D) both target (0,0) — the carrier wins the square but the ball still goes to the defender either way (verified below) — while D and R never target the same cell as player 1 at all.

Type: pure saddle, $V = +0.000000$ — player 0's maximin ties at D/R (not unique); player 1's minimax is uniquely R.

Solver's own policy (tie-broken): $p_0 = D \ 100\% \cdot p_1 = R \ 100\%$

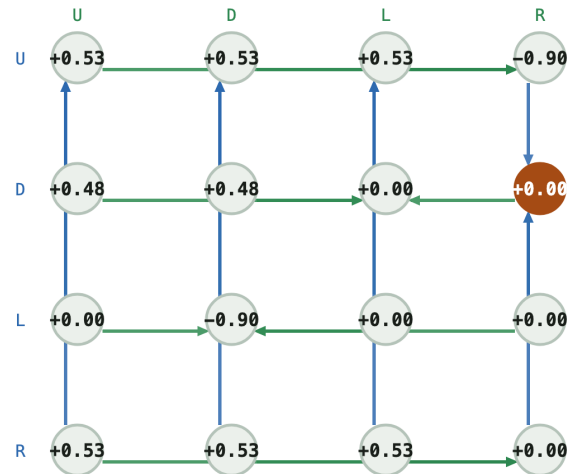
Display policy (uniform over ties, what the site shows and plays): $p_0 = D \ 50\% / R \ 50\% \cdot p_1 = R \ 100\%$

The tightest tie



pure equilibrium · $V = +0.000$

Q matrix -- (1, 0, 0, 1, 0)



pure saddle at D/R

Left: player 0's hold and one real arrow, 50/50. Right: the full 4×4 Q matrix as a node-and-arrow graph, correctly labelled a pure saddle at D/R despite the tie on both axes.

	U	D	L	R
U	0.531441	0.531441	0.531441	-0.900000
D	0.478297	0.478297	0.000000	0.000000
L	0.000000	-0.900000	0.000000	0.000000
R	0.531441	0.531441	0.531441	0.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x0, y0, x1, y1, b).

P0 \ P1	U	D	L	R
U	(1,1,0,2,0)	(1,1,0,0,0)	(1,1,0,1,0)	(1,1,0,1,1) – open contest, ball to loser
D	(1,0,0,2,0)	(1,0,0,0,0)	(1,0,0,1,0)	(1,0,1,1,0)
L	(0,0,0,2,0)	(0,0,0,1,1) – open contest, ball to loser	(0,0,0,1,0)	(0,0,1,1,0)
R	(2,0,0,2,0)	(2,0,0,0,0)	(2,0,0,1,0)	(2,0,1,1,0)

Case 8 (0, 0, 2, 0, 0) – The dead zone

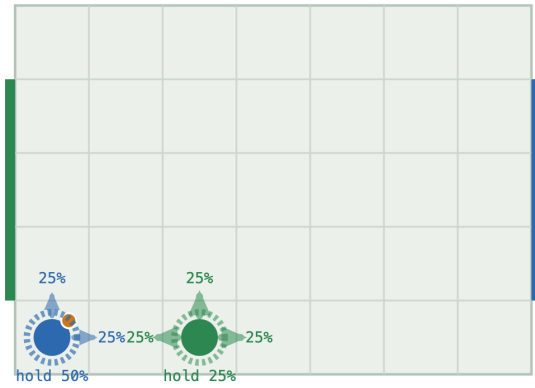
The single most common shape on the whole board (514 of 2,380 states): both players are fully indifferent among all four actions. It is not merely that the *guaranteed* values tie — every one of the 16 cells in the Q matrix is exactly 0.0, including a genuine ball-swapping contest ((R, L) → (1,0,2,0,1), verified below, itself also worth exactly 0). Nothing that happens this turn, for either player, changes anything.

Type: pure saddle, $V = +0.000000$ — both players' guarantees tie at all four actions (fully indifferent, and the entire matrix is flat zero).

Solver's own policy (tie-broken): $p_0 = U \ 100\% \cdot p_1 = U \ 100\%$

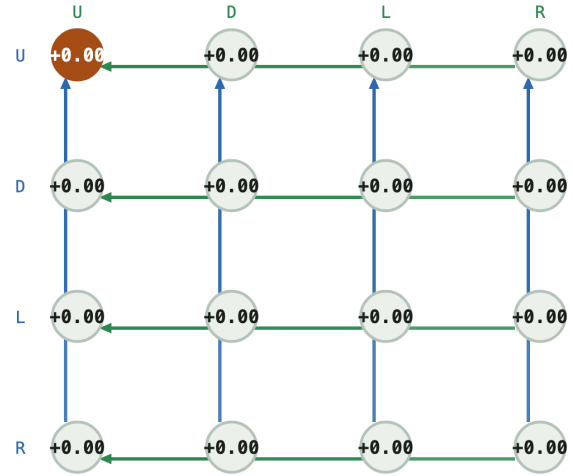
Display policy (uniform over ties, what the site shows and plays): $p_0 = U \ 25\% / D \ 25\% / L \ 25\% / R \ 25\%$
 $\cdot p_1 = U \ 25\% / D \ 25\% / L \ 25\% / R \ 25\%$

The dead zone



pure equilibrium · $V = +0.000$

Q matrix -- (0, 0, 2, 0, 0)



Left: both players' four equal-weight actions, two of player 0's wall-clamped into one hold ring. Right: a completely flat 4x4 Q matrix -- every cell is worth exactly 0.0.

	U	D	L	R
U	0.000000	0.000000	0.000000	0.000000
D	0.000000	0.000000	0.000000	0.000000
L	0.000000	0.000000	0.000000	0.000000
R	0.000000	0.000000	0.000000	0.000000

Successor states, by joint action — rows = player 0, columns = player 1, each cell the resulting (x_0, y_0, x_1, y_1, b) .

P0 \ P1	U	D	L	R
U	(0,1,2,1,0)	(0,1,2,0,0)	(0,1,1,0,0)	(0,1,3,0,0)
D	(0,0,2,1,0)	(0,0,2,0,0) — both hold	(0,0,1,0,0)	(0,0,3,0,0)
L	(0,0,2,1,0)	(0,0,2,0,0)	(0,0,1,0,0)	(0,0,3,0,0)
R	(1,0,2,1,0)	(1,0,2,0,0)	(1,0,2,0,1) — open contest, ball to loser (worth the same)	(1,0,3,0,0)

What these eight add up to

Every one of the 2,380 legal states on this board resolves to a pure saddle — the project's own headline deterministic-game result. Not one of them, however, has a fully unique best action for *both* players at once (Case 7's own

point): some tie is always present somewhere. These eight cases are not exceptions to "pure"; they are places where "pure" alone would have quietly understated the answer: a three-way tie displayed as a single arbitrary action, a wall-clamp displayed identically whether it holds or scores, a collision rule (the loser keeps the ball, win or lose the square) that is easy to get backwards without reading `game.transitions()` directly, and a stage game that can be a completely flat, inert 4×4 grid of zeros and still get reported as "arrows cycle, must mix" by a saddle-detector that breaks under a tie on both axes (Case 7 and 8 both caught this; fixed in `soccer_nash/viz.py`). Every number, tie set, and transition above is reproducible and independently checkable exactly as printed — see the footer.

Reproduce with `python scripts/a10_cases.py` (make `a10-cases`) — writes `figures/gallery/a10_cases.svg` and prints every exact matrix (via `NashQIteration.run_exact()`'s direct linear solve, not the iterative approximation to it), the full tied set, the solver's own tie-broken policy, the uniform-over-ties display policy, and the successor-state table for every joint action shown here, read straight from `game.transitions()`. The hold vs. score distinction and the uniform-over-ties board arrows are `soccer_nash/viz.py`'s `_wall_mask/_action_fan/policy_svg`; the Q-matrix graph's saddle detection (broken by a tie on both axes, caught building Cases 7 and 8) is `bestresponse_graph_svg`, fixed to check `maximin == minimax` directly instead of relying on two independent tie-breaks to agree. Full project: github.com/Charith-Reddy-Pareddy/soccer-markov-nash.